

## $M^5$ Day 9: Dividing polynomials

Name: \_\_\_\_\_

1. Divide these polynomials:

a. Divide  $x^3 + 2x + 1$  by  $x - 2$ .

b. Divide  $x^3 + x^2 + x^1 + 1$  by  $x + 1$ .

c. Divide  $3x^4 - 1$  by  $x^2 + 1$ .

**2.** Give an example of a degree 3 polynomial with roots  $-2$ ,  $2$  and  $3$ . (Leaving your answer in factored form is fine.) Is this unique?

**3.** Give an example of a degree 9 polynomial with roots  $-2$ ,  $2$  and  $3$  and only these roots. (Leaving your answer in factored form is fine.) Is this unique?

**4.** Give an example of a degree 4 polynomial that has no real roots.